

Program : Diploma in Architecture	
Course Code : 6181A	Course Title: Disaster Resilient Buildings
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To provide an overview of the occurrence, causes and consequences of disaster
- To understand of fundamental concepts and application of disaster resilient design.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Building Construction		Building Construction and Materials	3
Structural properties of RCC and Steel members		Design of Steel and concrete	5

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Discuss the hazards caused due to natural disaster and provide a brief insight to disaster mitigation and management.	10	Understanding
CO2	Identify the causes, impact and performance of structures, retrofitting and strengthening of existing structures both for cyclone and earthquake exclusively	16	Understanding
CO3	Explain with basic principles, simulation techniques, design considerations, adaptable building construction techniques	15	Understanding
CO4	Identify codes and practices separately for cyclone and earthquake resilient buildings	17	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3				2		
CO3	3						
CO4	3				2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Discuss the hazards caused due to natural disaster and provide a brief insight to disaster mitigation and management.		
M1.01	Recall the effects of natural disasters	3	Remembering
M1.02	Explain type of natural hazards	3	Understanding
M1.03	Outline disaster management and mitigation methods	4	Understanding
Contents: Brief introduction to different types of natural disaster, Occurrence of disaster in different climatic and geographical regions, hazard (earthquake and cyclone) map of the world and India, Regulations for disaster risk reduction, Post disaster recovery and rehabilitation (socio-economic consequences) - case studies.			
CO2	Identify the causes, impact and performance of structures, retrofitting and strengthening of existing structures both for cyclone and earthquake exclusively		
M2.01	Explain the effects of cyclone on structures	3	Understanding
M2.02	Outline the retrofitting and strengthening methods for resisting wind effects	5	Understanding
M2.03	Explain the effects of earthquakes on structures and non-structural elements	3	Understanding
M2.04	Interpret Seismic retrofitting and seismic strengthening methods	5	Understanding
	Series Test -I	1	

Contents:

Cyclone effects, Storm surge, Floods, Landslides. Behaviour of structures in past cyclones and wind storms, case studies. Cyclonic retrofitting, strengthening of structures and adaptive sustainable reconstruction. General planning/design considerations under wind storms & cyclones; Wind effects on buildings, towers, glass panels etc, & wind resistant features in design. Code Provisions, design wind speed, pressure coefficients; Coastal zoning regulation for construction & reconstruction phase in the coastal areas, traditional construction techniques in coastal areas.

Earthquake effects – On ground, soil rupture, liquefaction, landslides. Performance of ground and building in past earthquakes: Behaviour of various types of buildings, structures, and collapse patterns; Seismic retrofitting- Weakness in existing buildings, aging, concepts in repair, restoration and seismic strengthening.

CO3	Explain with basic principles, design considerations, adaptable building construction techniques		
M3.01	Explain general planning /design considerations under wind storms and cyclones	5	Understanding
M3.02	Explain various types of cyclone resistant construction techniques	5	Understanding
M3.03	Discuss innovative methods prevailing in the area	5	Understanding

Contents:

General planning /design considerations under wind storms and cyclones - Various Types of construction details of: Foundations, soil stabilization, plinth fill, flooring, walls, openings, roofs, terraces, parapets,- under-ground - staircases and isolation of structures; innovative construction material and techniques-

CO4	Identify codes and practices separately for cyclone and earthquake resilient buildings		
M4.01	Explain general Planning and design consideration for earthquake resilient buildings	7	Understanding
M4.02	Outline seismic effects related building configuration	6	Understanding
M4.03	Summarize the construction techniques for seismic resistant buildings	6	Understanding
	Series Test – II	1	

Contents:

General Planning and design consideration; Building forms, horizontal and vertical eccentricities, mass and stiffness distribution, soft storey etc.; Seismic effects related building configuration. Plan and vertical irregularities, redundancy and setbacks. Various Types and Construction details of: Foundations, plinth fill, flooring, walls, openings, roofs, terraces, parapets, and boundary walls, under-ground - staircases and isolation of structures.

Text / Reference:

T/R	Book Title/Author
R1	Burby, R. J. (1998). Cooperating with Nature. Confronting Natural Hazards with Land-Use Planning for Sustainable Communities. Washington: Joseph Henry Press
R2	Dutta, S. C. and Mukhopadhyay, P. (2012). Improving Earthquakes and Cyclone Resistance of Structures: Guidelines for the Indian Subcontinent. TERI
R3	ICIMOD. (2007). Disaster Preparedness for Natural Hazards: Current Status in India. Kathmandu: ICIMOD
R4	Judy, L. B. (2012). Climate change, Disaster Risk and the urban poor – cities building resilience for a changing World. Washington DC: The World Bank
R5	Holmes, J. D. (2007). Wind Loading of Structures. 2nd Ed. Taylor & Francis.
R6	Reducing Disaster Risk: A Challenge for Development. New York: UNDP. 26. World Bank. (2009). Handbook for Reconstructing after Natural Disasters.
R7	Christopher, A. and Reitherman, R. (1982). Building configuration and Seismic Design. John Wiley & Sons Inc.
R8	Dyrbye, C. D., Dyrbye, C. and Dyrbye, C. (1997). Wind Loads on Structures. John Wiley.
R8	Pelling, M. (2003). The Vulnerability of Cities: Social Resilience & Natural Disaster. London: Earthscan.